POLITECNICO DI MILANO

High Performance Computing Engineering

Quantum Physics – June 24, 2024

Please justify each answer and use a clear handwriting. Define each symbol you employ. For each exercise, give the symbolic solution and substitute numerical values (if required) only at the end. Write your first name, family name, and personal code on each sheet in a readable way.

1. A free particle of mass m can move only along the x axis. At time $t = 0$ the particle wavefunction is $\varphi(x)$, with $\varphi(x)$ *even*: $\varphi(x) = \varphi(-x)$.

a) Demonstrate that the particle wavefunction $\varphi(x, t)$ remains even also for $t \neq 0$.

b) Evaluate the particle momentum and position expectation values at $t = 0$. Do these values change with time?

Now consider a second particle described, at $t = 0$, by the wavefunction $\psi(x) = \varphi(x)e^{iqx}$, with q real.

c) Evaluate the momentum and position expectation values of the second particle at $t = 0$. Do these values change with time?

d) Find the energy expectation value of the particle described by $\psi(x)$, assuming that $\langle\varphi(x)|\hat{H}|\varphi(x)\rangle = E_0$. Does the obtained value change with time?

2. Consider two electrons in a two-level system characterized by the single particle energy eigenstates $|0\rangle$ and $|1\rangle$, with energy eigenvalues equal to E_0 and $E_1 > E_0$, respectively. The interaction between the electrons can be neglected.

a) Find the ground state ψ_g of the two-particle system and express it explicitly as a function of $|0\rangle$, $|1\rangle$, and the spinor of each electron. What is the corresponding energy eigenvalue?

b) When the system is in the ground state, find the expectation values of the squared module of the total spin and of the total spin component along a quantization axis of choice.

c) Determine the energy eigenstate(s) corresponding to the first excited level. What is the corresponding energy eigenvalue?

d) For each solution found at the previous point, find the expectation value of the total spin projection along a quantization axis of choice.

Solution to the 1st exercise

a) First, we note that the parity operator $\hat{\mathcal{P}}$, defined as $\hat{\mathcal{P}}u(x) = u(-x)$, *commutes* with the free particle Hamiltonian. Therefore, we can conclude from the Ehrenfest theorem that the parity $\langle\hat{\mathcal{P}}\rangle$ of a wavefunction is a conserved quantity:

$$\frac{d\langle\mathcal{P}\rangle}{dt} = \frac{1}{i\hbar}\langle[\hat{\mathcal{P}}, \hat{H}]\rangle = 0.$$

We can thus conclude that $\varphi(x,t)$ remains even, which means $\langle\varphi(x)|\hat{\mathcal{P}}|\varphi(x)\rangle = 1$, also for $t \neq 0$.

b) The position and momentum operators are *odd* with respect to the inversion of spatial coordinates, therefore, when the *even* wavefunction $\varphi(x)$ is considered, we obtain:

$$\langle x\rangle = \langle\varphi(x)|x|\varphi(x)\rangle = 0;$$
$$\langle p\rangle = \langle\varphi(x)|\hat{p}|\varphi(x)\rangle = 0.$$

Since $\varphi(x,t)$ remains even also for $t \neq 0$, these conclusions are true at any time.
The time-dependence of both $\langle x\rangle$ and $\langle p\rangle$ can be also evaluated by applying the Ehrenfest theorem. First, we notice that the momentum operator *commutes* with the free particle Hamiltonian, therefore momentum is a *conserved* quantity:

$$\frac{d\langle p\rangle}{dt} = \frac{1}{i\hbar}\langle[\hat{p}, \hat{H}]\rangle = 0 \qquad \Rightarrow \qquad \langle p(t)\rangle = \langle p(t=0)\rangle = 0.$$

Then, we evaluate the time dependence of $\langle x\rangle$:

$$\frac{d\langle x\rangle}{dt} = \frac{1}{i\hbar}\langle[x, \hat{H}]\rangle = \frac{1}{m}\langle p\rangle = 0 \qquad \Rightarrow \qquad \langle x(t)\rangle = \langle x(t=0)\rangle = 0.$$

The conclusion is that neither the momentum nor the position expectation value of the particle described at $t = 0$ by the particle wavefunction $\varphi(x)$ changes with time.

c) $\langle p\rangle = \langle\psi(x)|\hat{p}|\psi(x)\rangle = \hbar q;$
$\langle x\rangle = \langle\psi(x)|x|\psi(x)\rangle = \langle\varphi(x)|x|\varphi(x)\rangle = 0.$

As discussed in the previous point, the momentum of a free particle is a conserved quantity, therefore $\langle p\rangle$ does not change with time.
By applying the Ehrenfest theorem to $\langle x\rangle$, we obtain

$$\frac{d\langle x\rangle}{dt} = \frac{1}{i\hbar}\langle[x, \hat{H}]\rangle = \frac{1}{m}\langle p\rangle = \frac{\hbar q}{m} = v,$$

with $v =$ constant average particle velocity. Therefore, by integrating the previous expression with respect to time, we find:

$$\langle x(t)\rangle = vt + \langle x(t=0)\rangle = vt.$$

d) $\langle\psi(x)|\hat{H}|\psi(x)\rangle = E_0 + \frac{\hbar^2 q^2}{2m} = E_0 + \frac{1}{2}\frac{\langle p\rangle^2}{m} = E_0 + \frac{1}{2}mv^2.$

Solution to the 2nd exercise

a) $\psi_g = \frac{1}{\sqrt{2}}|0^{(1)}\rangle|0^{(2)}\rangle\left(|\uparrow^{(1)}\rangle|\downarrow^{(2)}\rangle - |\downarrow^{(1)}\rangle|\uparrow^{(2)}\rangle\right)$,

with $|\uparrow\rangle = \begin{bmatrix}1\\0\end{bmatrix}$ and $|\downarrow\rangle = \begin{bmatrix}0\\1\end{bmatrix}$ indicating the eigenstates of an electron spin projection along a generic axis z. The corresponding energy eigenvalue is $2E_0$.

b) $\hat{S}_z^{\text{tot}} = \hat{s}_z^{(1)} + \hat{s}_z^{(2)} \quad \Rightarrow \quad \hat{S}_z^{\text{tot}}\psi_g = 0$

$\Rightarrow \quad \langle\hat{S}_z^{\text{tot}}\rangle = 0$, with 0 being an eigenvalue of $\hat{S}_z^{\text{tot}}$.

$$\left(\hat{\mathbf{S}}^{\text{tot}}\right)^2 = \left[\hat{\mathbf{s}}^{(1)} + \hat{\mathbf{s}}^{(2)}\right]^2 = \left[\hat{\mathbf{s}}^{(1)}\right]^2 + \left[\hat{\mathbf{s}}^{(2)}\right]^2 + 2\hat{\mathbf{s}}^{(1)}\cdot\hat{\mathbf{s}}^{(2)}$$
$$= \left[\hat{\mathbf{s}}^{(1)}\right]^2 + \left[\hat{\mathbf{s}}^{(2)}\right]^2 + 2\left[\hat{s}_x^{(1)}\hat{s}_x^{(2)} + \hat{s}_y^{(1)}\hat{s}_y^{(2)} + \hat{s}_z^{(1)}\hat{s}_z^{(2)}\right]$$

$$\left[\hat{\mathbf{s}}^{(1)}\right]^2\psi_g = \left[\hat{\mathbf{s}}^{(2)}\right]^2\psi_g = \frac{3}{4}\hbar^2\psi_g$$
$$\hat{s}_x^{(1)}\hat{s}_x^{(2)}\psi_g = \hat{s}_y^{(1)}\hat{s}_y^{(2)}\psi_g = \hat{s}_z^{(1)}\hat{s}_z^{(2)}\psi_g = -\frac{1}{4}\hbar^2\psi_g \qquad \Rightarrow \qquad \left(\hat{\mathbf{S}}^{\text{tot}}\right)^2\psi_g = 0$$

c) There are four orthogonal eigenstates for the first excited level that are consistent with the spin statistics theorem:

$$\psi_a = \frac{1}{\sqrt{2}}\left(|1^{(1)}\rangle|0^{(2)}\rangle - |0^{(1)}\rangle|1^{(2)}\rangle\right)|\uparrow^{(1)}\rangle|\uparrow^{(2)}\rangle$$

$$\psi_b = \frac{1}{\sqrt{2}}\left(|1^{(1)}\rangle|0^{(2)}\rangle - |0^{(1)}\rangle|1^{(2)}\rangle\right)|\downarrow^{(1)}\rangle|\downarrow^{(2)}\rangle$$

$$\psi_c = \frac{1}{2}\left(|1^{(1)}\rangle|0^{(2)}\rangle - |0^{(1)}\rangle|1^{(2)}\rangle\right)\left(|\uparrow^{(1)}\rangle|\downarrow^{(2)}\rangle + |\downarrow^{(1)}\rangle|\uparrow^{(2)}\rangle\right)$$

$$\psi_d = \frac{1}{2}\left(|1^{(1)}\rangle|0^{(2)}\rangle + |0^{(1)}\rangle|1^{(2)}\rangle\right)\left(|\uparrow^{(1)}\rangle|\downarrow^{(2)}\rangle - |\downarrow^{(1)}\rangle|\uparrow^{(2)}\rangle\right)$$

The corresponding energy eigenvalue is $E_0 + E_1$.

d) The kets found at the previous point are also eigenstates of the total spin momentum projected along the quantization axis z:

$$\hat{S}_z^{\text{tot}}\psi_a = \hbar\psi_a \quad \Rightarrow \quad \langle\psi_a|\hat{S}_z^{\text{tot}}|\psi_a\rangle = \hbar$$
$$\hat{S}_z^{\text{tot}}\psi_b = -\hbar\psi_b \quad \Rightarrow \quad \langle\psi_b|\hat{S}_z^{\text{tot}}|\psi_b\rangle = -\hbar$$
$$\hat{S}_z^{\text{tot}}\psi_c = 0 \quad \Rightarrow \quad \langle\psi_c|\hat{S}_z^{\text{tot}}|\psi_c\rangle = 0$$
$$\hat{S}_z^{\text{tot}}\psi_d = 0 \quad \Rightarrow \quad \langle\psi_d|\hat{S}_z^{\text{tot}}|\psi_d\rangle = 0$$